Gather

$$P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f \tag{1}$$

$$x^2 + x = 10 \tag{2}$$

Multiline

$$\begin{aligned}(a+b+c+d)x^5 + (b+c+d+e)x^4 \\ + (c+d+e+f)x^3 + (d+e+f+a)x^2 + (e+f+a+b)x \\ + (f+a+b+c)\end{aligned}$$